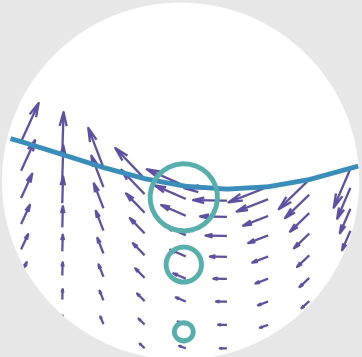

OCEAN WAVES AND TIDES

OE 754/854

Instructor Info

Prof. Nathan Laxague

MW 9:30 am - 10:30 am

Chase 121D

603-862-4023

http://unh-cassll.github.io

Nathan.Laxague@unh.edu

Lecture Info

Prereq: PHYS 407; MATH 527 & 528 *or* MATH 525 & 526

Monday, Wednesday, & Friday

8:10 am - 9:00 am

Kingsbury N343

Text Info

Dean & Dalrymple *Water Wave Mechanics for Engineers and Scientists*, 2nd Ed., World Scientific, 1991, ISBN 978-9810204211

Overview

OE 754/854 is an upper division undergraduate/first year graduate-level course on wave and tidal theory for engineers and scientists.

Wave-related topics include: Small amplitude wave theory, standing and propagating waves, transformation in shallow water, energy, random seas, forces on structures, and scaling of physical models in wave tank experiments.

Tide-related topics include: Tidal forcing and tidal dynamics. Description of tides in the ocean, tidal generating forces, equilibrium tide, dynamics of tides in coastal waters, and an introduction to the mathematical basis of tidal analysis.

Examples are taken from UNH research in the Chase wave tank and field observations in the Gulf of Maine.

Material

Required Text

Dean & Dalrymple, *Water Wave Mechanics for Engineers and Scientists*, 2nd Ed., World Scientific, 1991, ISBN 978-9810204211

This textbook is available in paperback and is plentiful, having been reprinted many times since its first printing in 1991. You should buy one and keep it nearby.

Other

Any other required reading materials will be provided through Canvas. Weekly computational notebooks accompany the lectures and are the fastest way to see the theory produce numbers.

Class Attendance & Preparation

You will be expected to attend class prepared for the lecture and ready to take notes for yourself. Often, this will include reading the textbook as assigned from one class to the next. If you need to miss class for a planned University-related activity, please let me know ahead of time. In the event that you need accommodation for a religious or cultural holiday/observance, you are encouraged to make that request as early in the semester as possible.

Three meetings this term—Friday, September 25th, Friday, October 2nd, and Wednesday, October 14th—will be delivered as recorded lectures posted to Canvas, paired with that week's notebook and reading. They cover required material and are not optional.

Grading Scheme

12%	Homework Assignments (x6, credit for submission)
48%	Midterm Exams (24% per exam)
40%	Final Exam

	A	B	C	D	F
+		87-89	77-79	67-69	
	93-100	83-86	73-76	63-66	<60
-	90-92	80-82	70-72	60-62	

Graded Assignments

- *Homework is graded for submission, not for content.* Each of the six assignments is worth one point, awarded for a good-faith attempt turned in on time; six points is 12% of the course grade. I am not marking your algebra. Homework will be assigned at the end of lecture and due (via upload through Canvas) as listed in the schedule.

- A fully worked solution set is released within two days of each deadline. The point of a problem set here is to find out which parts of the theory you cannot yet reproduce on your own. Eighty-eight percent of the grade rides on the three exams, where the same skills are tested under conditions in which you are time and tool-limited. Skip the homework assignments at your own risk.

- Midterm and Final exams will be administered in person, in Kingsbury N343. An equation sheet is provided with each exam; you do not need to prepare your own.

Exam Scheduling

Exams will be proctored in Kingsbury N343 during marked exam dates/times. It is my goal to create a learning experience that is as accessible as possible. If you anticipate any issues related to the testing requirements of this course or need accommodations, please either discuss them directly with me or in conjunction with the Student Accessibility Services Office within the first week of classes to explore alternative options.

Academic Integrity

You are required to comply with all University policies regarding Academic Integrity: https://catalog.unh.edu/srrr/student-policies-regulations/academic-integrity/. Do honest work; anything else deprives yourself of a learning opportunity that you only have for a short time.

Because homework is graded only for submission, there is no grade to be gained by copying it—only an exam you are less ready for. Use whatever resources help you learn the material, then close them and check whether you can still do the problem.

Use of Generative AI Tools

Apropos to the sections on graded assignments and academic integrity... Generative AI tools are powerful. You are presumably taking this class to learn about ocean waves and improve your ability to solve problems. So, should you wish to use these tools during the semester, make sure that you're building a durable knowledge base and working to strengthen your problem-solving abilities. <u>You will not have access to AI tools during exam time.</u> You will likely have access to them when you are gainfully employed, but so will everyone else in your field. Don't deprive yourself of the opportunity to become better <u>now</u>.

Conduct and Respect for Peers

All participants in OE 754/854 (including myself and the students) shall treat each other with respect and collegiality. We endeavor to create a welcoming, friendly, and inclusive environment for everyone. To do otherwise marginalizes individuals who are here to learn and grow. Participation is of great importance to an intellectually vibrant class experience. To this end, in order to ensure a climate of learning for all, disruptive or inappropriate behavior (repeated outbursts, disrespect for others, etc.) may result in exclusion (removal) from this class.

Accommodations for Students with Disabilities

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, please contact Student Accessibility Services (SAS) at 201 Smith Hall. If you have received an accommodation letter for this class, please contact me immediately so we can discuss the necessary arrangements. SAS may be contacted at https://www.unh.edu/student-accessibility-services,
(603) 862-2607, sas.office@unh.edu.

Lecture Schedule

WEEK	DATES	OVERALL TOPIC	CHAPTER
Week 1	Aug 31 – Sep 2	Introduction & ocean wave phenomena	Chap. 1
Week 2	Sep 4 – 11	Fluid mechanics review & governing equations	Chap. 2
Week 3	Sep 14 – 18	Boundary conditions & BVP formulation	Chap. 2–3
Week 4	Sep 21 – 25	Linearization & the Airy solution	Chap. 3
Week 5	Sep 28 – Oct 2	Wave kinematics: particle motion & pressure	Chap. 3
Week 6	Oct 5 – 9	Wave energy, flux & group velocity	Chap. 4
Week 7	Oct 14 – 19	Wave transformation I: shoaling & refraction	Chap. 4
Week 8	Oct 21 – 23	Wave transformation II: diffraction, reflection, breaking	Chap. 4
Week 9	Oct 26 – 30	Finite-amplitude theories I: Stokes expansion	Chap. 11
Week 10	Nov 2 – 6	Finite-amplitude theories II: cnoidal & solitary	Chap. 11
Week 11	Nov 9 – 13	Tides	Chap. 5
Week 12	Nov 16 – 20	Forces on structures, wave setup, longshore currents	Chap. 5
Week 13	Nov 23 – 25	Random seas I: time-domain analysis	Chap. 7
Week 14	Nov 30 – Dec 4	Random seas II: spectral analysis	Chap. 7
Week 15	Dec 7 – 14	Directional spectra & modern wave observation	Notes

No class Monday, September 7th (Labor Day), Monday, October 12th (Mid-Semester Break), Wednesday, November 11th (Veterans Day), or Friday, November 27th (Thanksgiving recess). In place of in-person lectures, you will have access to recorded content and other asynchronous material on the following dates: Friday, September 25th, Friday, October 2nd, and Wednesday, October 14th.

Assignment/Exam Schedule

ASSIGNMENT/EXAM	ASSIGNED	DUE	TIME & LOCATION
Homework 01 Vector calculus, continuity, material derivative	Wed, Sep 9	Fri, Sep 18	Canvas upload
Homework 02 Boundary conditions, BVP, linearization	Fri, Sep 18	Fri, Oct 2	Canvas upload
Exam 01	–	Thu, Oct 8	12:40 PM - 2:00 PM, KING N343
Homework 03 Energy, group velocity, shoaling, refraction	Fri, Oct 9	Fri, Oct 23	Canvas upload
Homework 04 Stokes expansion, cnoidal/solitary, Ursell	Mon, Oct 26	Mon, Nov 2	Canvas upload
Exam 02	–	Thu, Nov 5	12:40 PM - 2:00 PM, KING N343
Homework 05 Morison, Sainflou, radiation stress, setup	Mon, Nov 16	Wed, Nov 25	Canvas upload
Homework 06 Spectra, moments, random sea synthesis	Mon, Nov 30	Fri, Dec 11	Canvas upload
Final Exam	–	Mon, Dec 21	1:00 PM - 3:00 PM, KING N343

Exam 01 covers the material of Homeworks 01–02 (Weeks 1–5); Exam 02 covers Homeworks 03–04 (Weeks 6–10); the Final Exam is cumulative, with emphasis on the topics of Homeworks 02–05. Tides, random seas, and spectral analysis are examined on the Final Exam only. Review sessions are held in the class meeting immediately before each midterm (Wed, Oct 7 and Wed, Nov 4).

Graduate Student (OE 854) Term Paper

In order to receive credit for OE 854—the graduate-level version of the course—students are expected to complete a term paper in addition to satisfying the other course requirements. This paper should constitute a review of the scholarly literature relevant to one of the topics covered during the semester. Seed publications for a handful of these topics are provided in the table below. Settle your topic with me by the end of Week 7. While there is no formal length requirement, for a paper to receive credit it should be lengthy enough to contain a thoughtful review of the literature.

SUBJECT	CITATION
Water Wave Theory	Craik, A. D. (2004). The origins of water wave theory. *Annual Review of Fluid Mechanics*, 36(1), 1-28.
Wave-Current Interaction	Wolf, J., and Prandle, D. (1999). Some observations of wave–current interaction. *Coastal Engineering*, 37(3-4), 471-485.
Engineering Wave Properties	Peregrine, D. H. (1983). Breaking waves on beaches. *Annual Review of Fluid Mechanics*, 15(1), 149-178.
Tides and Estuarine Physics	MacCready, P., and Geyer, W. R. (2010). Advances in estuarine physics. *Annual Review of Marine Science*, 2(1), 35-58.
Stokes Drift	van den Bremer, T. S., and Breivik, Ø. (2018). Stokes drift. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 376(2111), 20170104.